Christian Olivares-Rodriguez

Machine Learning Engineer | AI Systems & Data Science | MS Artificial Intelligence (May 2027)
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TECHNICAL SKILLS

- **Languages:** Python, SQL, TypeScript, Next.js
- **Data Science & ML:** Pandas, NumPy, PySpark, Scikit-Learn, Jupyter, Pytorch, GeoPandas, Artificial Neural Networks
- **Infrastructure & Ops:** Linux (Ubuntu), SSH, Bash Scripting, GPU Provisioning (CUDA), Tailscale/VPN
- **Visualization & Cloud:** Tableau, Matplotlib, Plotly, Folium , Google Cloud Platform (GCP Dataproc), AWS S3
- **Competencies:** Data Cleaning, EDA, ETL, Predictive Modeling, Root Cause Analysis

EDUCATION

Arizona State University | Tempe, AZ | **M.S. Artificial Intelligence** (*May 2027*) | **B.S. Data Science** (*Magna Cum Laude*, May 2025)

PROFESSIONAL EXPERIENCE

CO³ Labs | **Founder & AI Engineer** | *September 2025 – Present*

- **Custom AI Agent Architecture:** Designing and developing a modular agent system in **Python**, focused on **context management**, **task execution**, and autonomous decision-making workflows.
- **Local LLM &Low-Level Workflows:** Building autonomous agent logic using local LLMs with fine-grained execution controls, avoiding high-level framework dependencies for maximum transparency and flexibility.
- **GPU Compute Infrastructure:** Architected and deployed a headless **Linux compute node** (**RTX 4090**) for AI development, enabling remote access via **SSH** and secure **mesh networking**.

Geek Squad | **Consultation Agent & AutoTech** | *January 2019 – Present*

- **Orchestrated Technical Operations:** Managed the lifecycle of complex hardware and software repair tickets, prioritizing tasks in a high-volume environment to ensure service level agreements (SLAs) were consistently met.
- **Systematic Root Cause Analysis:** Executed advanced diagnostic protocols to isolate and resolve system failures, utilizing logic-based troubleshooting methodologies that parallel software debugging.
- **Technical Communication:** Translated complex diagnostic results into actionable insights for non-technical clients, building trust and delivering data-backed hardware solutions.

The Global Career Accelerator | **Data Analytics Project Intern**| *January 2025 – May 2025*

- **Key Industry Projects:** Selected for **Intel** (Sustainability) and **The Recording Academy (GRAMMYs)** (Audience Analysis) tracks, utilizing **Python** and **SQL** to analyze real-world datasets and drive strategy for global brands.
- **Technical Specialization:** Earned **Python** **&** **SQL Specialist** badges, mastering advanced data manipulation in **Jupyter** (Pandas/Plotly) and complex querying (JOINs/Aggregations) to solve business problems.

TECHNICAL PROJECTS & HACKATHONS

Chicago Crime Project | *Remote* | *2026 – Present*

- Designed and trained a **4-layer PyTorch neural network** on 192,082 Chicago crime records to predict arrest likelihood, achieving **86% accuracy** with weighted sampling to handle a 75/25 class imbalance across 33 crime categories.
- Engineered a **60-feature input matrix** combining one-hot encoded crime types, geospatial district/beat identifiers, and temporal features; evaluated model performance via precision, recall, and F1-score achieving **0.78 arrest precision** and **0.91 no-arrest F1**.

Ookla Broadband Equity Analysis | *Remote* | *2026 – Present*

- **Engineered** a scalable data pipeline to extract and process 400K+ real-world speed test tiles from AWS S3 (Ookla Open Data), applying geospatial bounding box filtering with RAM-efficient chunked processing.
- **Conducted** socioeconomic equity analysis via geospatial joins (GeoPandas) with U.S. Census data, discovering that rural classification is a 3x stronger predictor of broadband gaps than median income.

Vercel x Nvidia AI Hackathon | *San Francisco, CA* | *October 2025*

- Built a full-stack **Next.js** web application on **Vercel Edge Runtime** that accepts natural language dataset descriptions and returns AI-generated schemas and downloadable CSV samples via a /api/generate endpoint.
- Integrated **NVIDIA Nemotron** with automatic JSON repair logic and preset templates for retail, IoT, and support ticket datasets to address ML data scarcity.

Fraud Detection on GCP with PySpark | *Remote* | *December 2024*

- Deployed **PySpark** on **GCP Dataproc** to process 999,001 transaction records; engineered features via **StringIndexer** and trained **Logistic Regression** and **Random Forest** classifiers using **PySpark MLlib**, evaluating F1, precision, and recall.